Marvyn Bailly

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SUMMARY

Applied Mathematics Ph.D. candidate in mesh generation, computational geometry, numerical optimization, and numerical PDEs, with ML/AI project work. Builds tested, documented, user-facing software in Julia, Python, and C++.

EDUCATION

University of Maryland: PhD, Applied Mathematics & Scientific Computation (GPA 3.89) *2023 - Present*
University of Washington: MS, Applied Mathematics, Numerical Methods (GPA 3.84) *2022 - 2023*
Oregon State University: BS, Mathematics, Applied & Computational (GPA 3.96) *2020 - 2022*

Relevant coursework: *Numerical Methods for Partial Differential Equations · Numerical Optimization · High-Performance Computing · Natural Language Processing*

EXPERIENCE

Graduate Research Assistant, University of Maryland *College Park, MD*
Computational Turbulence Lab (Advisor: Dr. Johan Larsson) *2024 - Present*

- Designed mesh generation algorithms for CFD (multi-block structured and metric-driven quad meshes) using computational geometry, discrete differential geometry, optimization, and PDE methods.
- Built them into tested, documented libraries with a GUI, cutting mesh setup from days of manual work to hours of automatic computation.
- Presented methods to multidisciplinary teams, turning simulation needs into optimization constraints.

Atmospheric & Oceanic Science (Advisor: Dr. Jacob Wenegrat) *2023 - 2024*

- Extended a published ocean-convection study to a more realistic model built in Julia, running large eddy simulations on an HPC cluster.

Freelance Software Developer *for N. Warrington (MIT) & S. Sorokanich (AFRL), remote* *Apr 2026 - Jul 2026*

- Built Quantum Circuit Builder (JavaScript), a web app turning superconducting circuit diagrams into Hamiltonian parameters for lattice field theory, replacing hand derivation.

TEACHING & LEADERSHIP

Teaching Assistant, University of Maryland & University of Washington *2022 - Present*

- Led weekly discussion sections in graduate and undergraduate data science, probability, and calculus.

Summer Camp Director, Girls Talk Math, University of Maryland *2023 - 2025*

- Co-directed the camp: ran operations, hired student leaders, and coordinated industry outreach.

Graduate Student Committee, University of Maryland *2025 - 2026*

- Co-organized and led the weekly Joint AMSC, Math & Stat Student Seminar (JAMS).

SELECTED PROJECTS

- **MetricRSP** (Python) — automatic quad mesher (metric field in, CFD-ready mesh out) using Delaunay triangulation, cross fields, L-BFGS/Newton fold-free parameterization, and integer-grid quantization.
- **GridGeneration.jl** (Julia) — semi-automatic multi-block grid generator using algebraic and PDE-driven methods; ships generated API documentation and worked airfoil examples.
- **Smart Shelf Auditor** (Python) — team-built AI agent that audits retail shelves with object detection and a vision-language model; Finalist, UMD Agentic AI Challenge (top 5 of 67 teams, Apr 2026).
- **Mechanistic Interpretability of Grammatical Structures** (Python, PyTorch) — path patching across GPT-2 small's 12 layers and 144 attention heads to locate a name-resolution circuit.

TECHNICAL SKILLS

- **Programming:** Julia, Python, JavaScript, C++, MATLAB, Java.
- **Numerics & CFD:** finite volume/element methods, large eddy simulation, numerical optimization, SU2, Gmsh.
- **Tools:** Linux, Git, Make, SLURM, MPI, OpenMP, NumPy/SciPy, PyTorch, LaTeX.